The volatility of the costs of raw materials has impacted our operations over the past several years. We, and others in our industry, generally have been able to pass cost increases on major raw materials through to our customers using surcharges that are structured to recover increases in raw material costs. Generally, the formula used to calculate a surcharge is based on published prices of the respective raw materials for the previous month which correlates to the prices we pay for our raw material purchases. However, a portion of our surcharges to customers may be calculated using a different surcharge formula or may be based on the raw material prices at the time of order, which creates a lag between surcharge revenue and corresponding raw material costs recognized in cost of sales. The surcharge mechanism protects our net income on such sales except for the lag effect discussed above. However, surcharges have had a dilutive effect on our gross margin and operating margin percentages as described later in this report.

Approximately 25 percent of our net sales are sales to customers under firm price sales arrangements. Firm price sales arrangements involve a risk of profit margin fluctuations, particularly when raw material prices are volatile. In order to reduce the risk of fluctuating profit margins on these sales, we enter into commodity forward contracts to purchase certain critical raw materials necessary to produce the related products sold. Firm price sales arrangements generally include certain annual purchasing commitments and consumption schedules agreed to by the customers at selling prices based on raw material prices at the time the arrangements are established. If a customer fails to meet the volume commitments (or the consumption schedule deviates from the agreed-upon terms of the firm price sales arrangements), the Company may need to absorb the gains or losses associated with the commodity forward contracts on a temporary basis. Gains or losses associated with commodity forward contracts are reclassified to earnings/loss when earnings are impacted by the hedged transaction. Because we value most of our inventory under the LIFO costing methodology, changes in the cost of raw materials and production activities are recognized in cost of sales in the current period attempting to match the most recently incurred costs with revenues. Gains and/or losses on the commodity forward contracts are reclassified from other comprehensive income together with the actual purchase price of the underlying commodities when the underlying commodities are purchased and recorded in inventory. To the extent that the total purchase price of the commodities, inclusive of the gains or losses on the commodity forward contracts, are higher or lower relative to the beginning of year costs, our cost of goods sold reflects such amounts. Accordingly, the gains and/or losses associated with commodity forward contracts may not impact the same period that the firm price sales arrangements revenue is recognized, and comparisons of gross profit from period to period may be impacted. These firm price sales arrangements are expected to continue as we look to strengthen our long-term customer relationships by expanding, renewing and, in certain cases, extending to a longer term, our customer long-term arrangements.

We produce hundreds of grades of materials, with a wide range of pricing and profit levels depending on the grade. In addition, our product mix within a period is subject to the fluctuating order patterns of our customers as well as decisions we may make on participation in certain products based on available capacity including the impacts of capacity commitments we may have under existing customer agreements. While we expect to see positive contribution from a more favorable product mix in our margin performance over time, the impact by period may fluctuate, and period to period comparisons may vary.

Net Pension Expense

Net pension expense, as we define it below, includes the net periodic benefit costs related to both our pension and other postretirement plans. The net periodic benefit costs are determined annually based on beginning of year balances and are recorded ratably throughout the fiscal year, unless a significant re-measurement event occurs. The following is a summary of the net periodic benefit costs for the years ended June 30, 2019, 2018 and 2017:

	Years Ended June 30,		
($ in millions)	**2019**	**2018**	**2017**
Pension plans	$ 9.8	$ 11.3	$ 45.8
Other postretirement plans	1.8	2.9	2.6
Net periodic benefit costs	$ 11.6	$ 14.2	$ 48.4

In September 2016, we announced changes to retirement plans we offer to certain employees. Benefits accrued to eligible participants of our largest qualified defined benefit pension plan and certain non-qualified pension plans were frozen effective December 31, 2016. Approximately 1,900 affected employees were transitioned to the Company's 401(k) plan that has been in effect for eligible employees since 2012, when the pension plan was closed to new entrants. We recognized the plan freeze during fiscal year 2017 as a curtailment, since it eliminated the accrual for a significant number of participants for all of their future services. We also made a voluntary pension contribution of $100.0 million to the affected plan in October 2016.

The service cost component of net pension expense represents the estimated cost of future pension liabilities earned associated with active employees. The pension earnings, interest and deferrals ("pension EID") is comprised of the expected return on plan assets, interest costs on the projected benefit obligations of the plans and amortization of actuarial gains and losses and prior service costs.

Net pension expense is recorded in accounts that are included in both the cost of sales and selling, general and administrative expenses based on the function of the associated employees and in other income (expense), net. The following is a summary of the classification of net pension expense for the years ended June 30, 2019, 2018 and 2017:

	Years Ended June 30,		
($ in millions)	**2019**	**2018**	**2017**
Cost of sales			
Service cost	$ 10.0	$ 10.5	$ 20.2
Total cost of sales	10.0	10.5	20.2
Selling, general and administrative expenses			
Service cost	1.5	1.6	3.9
Total selling, general and administrative expenses	1.5	1.6	3.9
Other expense			
Pension earnings, interest and deferrals	0.1	2.1	23.8
Curtailment charge	—	—	0.5
Total other expense	0.1	2.1	24.3
Net pension expense	$ 11.6	$ 14.2	$ 48.4

As of June 30, 2019 and 2018, amounts capitalized in gross inventory were $1.7 million and $1.7 million, respectively.

Operating Performance Overview

We believe fiscal year 2019 was a successful year as strong execution of our commercial and manufacturing strategies combined with growing market demand across most markets resulted in our best operating income performance since fiscal year 2013. We made significant progress in the following areas:

- We expanded customer relationships through market share gains with new and existing customers based on the strength of our solutions focused customer approach.
- Through the ongoing implementation of the Carpenter Operating Model, we have unlocked incremental capacity through efficiency and productivity improvements across our SAO and PEP businesses.
- Our Athens, Alabama facility continues to gain qualifications of submitted Aerospace Vendor Approved Processes contributing to increased capacity.
- We expanded our leading additive manufacturing platform with the acquisition of LPW Technology Ltd.
- We increased our focus and investment in core growth areas such as additive manufacturing and soft magnetics.

Results of Operations — Fiscal Year 2019 Compared to Fiscal Year 2018

For fiscal year 2019, we reported net income of $167.0 million, or $3.43 per diluted share. Excluding special items, earnings per share would have been $3.46 per diluted share for fiscal year 2019. This compares with net income of $188.5 million, or $3.92 per diluted share, a year earlier. Excluding special items, earnings per share would have been $2.50 per diluted share for fiscal year 2018. Our fiscal year 2019 results reflect strong market conditions combined with our solutions focused approach that drove increasing sales in the majority of our end-use markets and further implementation of the Carpenter Operating Model.